

Radiation Sensitivity

Ultraviolet radiation (MC1R, pigmentation, scalding, tanning and skin cancers), ionizing radiation (ATM, XRCC1, XRCC3, TGFB1), thyroid and radiation (FOXE1, NKX2-1) and radiotherapy toxicity

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. In radiation UV the profile is protective: none of the 7 variants of **MC1R** is present, IRF4 and TYR R402Q also do not have the risk alleles, and there is 1 protective allele in **SLC45A2**. The scalding in childhood are at percentile 0.8 (loci load at 3.2), melanoma at the 6.5 and 12.3 percentiles, squamous cell carcinoma at 3.4 and basal cell carcinoma in 0.9 (in an 2.0 score, at 69.8). In the repair of DNA, XRCC1, XRCC3 and ERCC2 have common genotypes.

To be considered. In ionizing radiation there are unfavourable signs: ATM D1853N in homozygosity (A/A), 1 allele of TGFBI -509 (fibrosis) and loci load of radiotherapy toxicity at percentile 90.0. In thyroid cancer, 1 risk allele in FOXE1 and NKX2-1, score at percentile 79.1 and loci load at 93.5. The ease in tanning is low to typical (percentile 20.9).

What does it mean? The skin tolerates the sun well from a genetic point of view, but isotretinoin increases photosensitivity while taken. Genetic sensitivity to ionizing radiation and thyroid is slightly higher than the average: it is worth avoiding TAC and X-rays without clear indication and knowing the home radon.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 22 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 20 of the chip, 2 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 5 SNPs needed conversion to the complementary strand and 3 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 8 PGS Catalog scores calculated locally with 1,318,760 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 841 loci, 735 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In scalding and cancer scores (melanoma, basal cell, squamous cell, thyroid), high percentile = higher risk. In the score of ease in tanning, high percentile = bronze more easily. There are no published polygenic scores for radiosensitivity.

Sun and skin

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Ease in bronze	PGS002155 (Privé 2022, LDpred2, 241 thousand variants)	237,411 240,701 (weighted coverage 99%)	20.9	TYPICAL	without judgment
Childhood Scales	PGS001257 (Tanigawa 2022, 519 variants)	444 519 (weighted coverage 94%)	0.8	LESS LIKELY	favourable

- **Tanning facility:** High percentile = bronze more easily (more natural protection).

Skin cancer

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Melanoma	PGS000079 (Graff 2021, 24 variants)	22 24 (weighted coverage 94%)	6.5	LESS LIKELY	favourable
Melanoma (genomic)	PGS000409 (Fritsche 2020, PRS-CS, 1.1 million variants)	1,080,379 1,111,490 (weighted coverage 96%)	12.3	LESS LIKELY	favourable
Basal cell carcinoma	PGS000119 (Fritsche 2019, 32 variants)	32 32 (weighted coverage 100%)	0.9	LESS LIKELY	favourable
Basal cell carcinoma	PGS003416 (462 variants (MTAG))	440 462 (weighted coverage 92%)	69.8	TYPICAL	typical
Skin squamous cell carcinoma	PGS005209 (20 variants)	20 20 (weighted coverage 100%)	3.4	LESS LIKELY	favourable

Thyroid

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Thyroid cancer	PGS000087 (Graff 2021, 12 variants)	12 12 (weighted coverage 100%)	79.1	TYPICAL	typical

4. Key variants by topic

Variants with described effect on skin response to UV, DNA repair after ionizing radiation and thyroid cancer.

MC1R (skin response to sun)

MC1R is the receptor that, after the sun, causes melanocytes to produce eumelanin (brown pigment, protector). The variants "R" (R151C, R160W, D294H and others) deviate the production to pheomelanin (red pigment), which protects little and generates free radicals with UV. Each R allele increases the risk of melanoma about 1.5–2 times, even without light skin.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
MC1R	rs1805007	R151C	C/C	chip	T (0 copy/s)	favourable	R151C ("R", red hair): less eumelanin, more scalds and more melanoma (OR ~1.5–2 per allele)	Strong
MC1R	rs1805008	R160W	C/C	chip	T (0 copy/s)	favourable	R160W ("R")	Strong
MC1R	rs1805009	D294H	G/G	chip	C (0 copy/s)	favourable	D294H ("R")	Strong
MC1R	rs1805005	V60L	G/G	chip	T (0 copy/s)	favourable	V60L ("r", minor effect)	Moderate
MC1R	rs11547464	R142H	G/G	chip	A (0 copy/s)	favourable	R142H ("R")	Moderate
MC1R	rs1805006	D84E	C/C	chip	A (0 copy/s)	favourable	D84E ("R")	Moderate
MC1R	rs1110400	I155T	T/T	chip	C (0 copy/s)	favourable	I155T ("R")	Moderate

Pigmentation and sensitivity to sun

The color of skin, eyes and hair depends on several genes (IRF4, SLC45A2, TYR, HERC2/OCA2, ASIP). Light skin, freckles and light eyes are associated with more scalding and skin cancer.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
IRF4	rs12203592		C/C	chip	T (0 copy/s)	favourable	IRF4: the T allele gives freckles, skin more sensitive to sun and less capacity to bronze	Strong
SLC45A2	rs16891982	L374F	C/G	chip	C (1 copy/s)	favourable	L374F: the C allele (Leu374) gives darker pigmentation and protects from melanoma; G (Phe374) gives lighter skin	Strong
TYR	rs1126809	R402Q	G/G	imputation DR2=1.00	A (0 copy/s)	favourable	R402Q (A = Gln402): thermolabile tyrosinase, lighter skin; associated with melanoma and basal cell carcinoma	Moderate
TYR	rs1042602	S192Y	C/A	chip	A (1 copy/s)	unfavourable (light)	S192Y (A = Tyr192): clearer pigmentation	Moderate
HERC2/OCA2	rs12913832		A/G	chip	G (1 copy/s)	unfavourable (light)	The main determinant of eye color (G = light eyes); light eyes are associated with more sensitivity to the sun	Moderate
ASIP	rs4911414	haplotype AH	G/G	chip	T (0 copy/s)	favourable	ASIP (MC1R antagonist): T is part of the haplotype of light skin, freckles and scalding associated with melanoma and basal cell carcinoma (tanning studies do not always agree in the direction)	Weak

DNA repair and ionizing radiation

The ionizing radiation (X, TAC, radiotherapy, radon) starts the DNA. ATM detects the breaks of the two chains, XRCC1 and XRCC3 repair them and TP53 decides between repairing and eliminating the cell. Variants of these genes are associated, in radiotherapy studies, with more late toxicity (fibrosis, skin, bladder, rectum), and TGF- β 1 is the fibrosis motor. ERCC2 mainly repairs the lesions of UV.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ATM	rs1801516	D1853N	A/A	chip	A (2 copy/s)	unfavourable	D1853N: associated in some studies with more late toxicity of radiotherapy	Weak
XRCC1	rs25487	R399Q	C/C	chip	A (0 copy/s)	favourable	R399Q (A gene = Gln399): repair by base excision; associated with radiosensitivity in some studies	Weak
XRCC1	rs1799782	R194W	G/G	chip	T (0 copy/s)	favourable	R194W (T of gene = Trp194)	Weak
XRCC3	rs861539	T241M	G/G	chip	T (0 copy/s)	favourable	T241M (gene T = Met241): homologous recombination	Weak
ERCC2 (XPD)	rs13181	K751Q	T/T	chip	C (0 copy/s)	favourable	K751Q (C of the gene = Gln751): nucleotide excision repair (UV injuries)	Weak
TP53	rs1042522	R72P	C/C	imputation DR2=0.97	G (0 copy/s)	without judgment	R72P (G in strand + = Pro72): less apoptosis and more cycle arrest after damage to DNA	Weak
TGFB1	rs1800469	-509	A/G	chip	T (1 copy/s)	unfavourable (light)	-509C>T (gene T): plus TGF- β 1; associated with fibrosis after radiotherapy	Weak

Thyroid and radiation

Thyroid is the most sensitive organ to ionizing radiation, especially in childhood (radioactive iodine, neck radiotherapy). FOXE1 and NKX2-1 are the strongest loci of thyroid cancer, both spontaneous and induced by radiation.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FOXE1	rs965513		A/G	chip	A (1 copy/s)	unfavourable (light)	FOXE1 (9q22): the strongest locus of thyroid cancer, including radiation-induced thyroid cancer after nuclear accidents [GWAS Catalog: Thyroid cancer, allele G, beta 0.519 [0.44-0.6] unit decrease, p=6E-40, PMID 36508215] (the reported G allele reduces the trait; here the effect allele is the opposite, A)	Strong
NKX2-1/MBIP	rs944289		C/T	chip	T (1 copy/s)	unfavourable (light)	14q13 (NKX2-1): locus of thyroid cancer, also in the form induced by radiation [GWAS Catalog: Thyroid cancer, T allele, OR 1.37, p=2E-9, PMID 19198613]	Moderate

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each trait, with each risk-oriented allele (in the difficulty in tanning, in the sense of tanning less), compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Scalds and sensitivity to the sun	42 (30)	35 of 60	30	3.2	favourable
Difficulty in tanning	29 (23)	19 of 46	23	61.1	typical
Melanoma	151 (143)	107 of 286	143	41.4	typical
Basal cell carcinomas and squamous cells	394 (361)	312 of 722	358	25.1	typical
Toxicity of radiotherapy	59 (40)	36 of 80	40	90.0	unfavourable
Thyroid cancer	166 (138)	137 of 276	138	93.5	unfavourable

6. Favourable

- **MC1R without variants:** R151C C/C, R160W C/C, D294H G/G, V60L G/G, R142H G/G, D84E C/C e I155T T/T. **Strong.**
- **Pigmentation:** IRF4 C/C without the allele of the freckles, TYR R402Q G/G absent, ASIP G/G without the allele of the clear skin haplotype, and SLC45A2 L374F C/G with protective 1 allele. **Moderate / strong.**
- **Low childhood scales** (percentile 0.8; loci load at 3.2). **Moderate.**
- **Skin cancer:** melanoma at the 6.5 and 12.3 percentiles (loci load at 41.4); squamous cell carcinoma at 3.4; basal cell carcinoma in 0.9 (the MTAG score gives 69.8; locus load of carcinomas at 25.1). **Moderate.**
- **DNA repair:** XRCC1 R399Q C/C and R194W G/G, XRCC3 T241M G/G and ERCC2 K751Q T/T with the common genotypes. **Weak.**

7. Unfavourable

- **ATM D1853N in homozygosity** (A/A): associated in radiotherapy meta-analysis with more late toxicity (OR ~1.2–1.5 by allele). It is not a causal mutation of ATM. **Weak / moderate.**
- **Toxicity of radiotherapy:** loci load at percentile 90.0; TGFB1 -509 A/G with 1 allele plus TGF- β 1 (fibrosis). **Weak.**
- **Thyroid cancer:** FOXE1 A/G and NKX2-1 C/T with 1 risk allele each; score at percentile 79.1 and loci load at 93.5. The absolute risk remains low. **Moderate.**
- **Pigmentation:** TYR S192Y C/A and HERC2/OCA2 A/G with 1 clearer pigmentation allele each; ease of tanning at percentile 20.9 (Loci burden of tanning difficulty at 61.1). **Weak.**
- **No judgment:** TP53 R72P C/C (Arg/Arg, the form with more apoptosis after damage in DNA).

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
DNA Damage and DNA Repair Deficit	UV: MC1R and excision repair (ERCC2) without variants; ionizing: ATM D1853N in homozygosity	Neutral
Genomic instability	XRCC1 and XRCC3 with common genotypes ATM Asn/Asn	Neutral
Somatic Mutation Accumulation	Low skin cancer in the scores (melanoma 6.5, squamous cell 3.4): less mutations of the accumulated UV	Favourable
Oxidative stress	Without excess pheomelanin (normal MC1R), less free radicals in the skin with UV	Favourable
Remodelling of the extracellular matrix	Photoaging: protective pigmentation profile; TGFB1 -509 with 1 allele profibrosis	Neutral
Thyroid alterations	Thyroid cancer: loci load at percentile 93.5, FOXE1 and NKX2-1 with 1 risk allele	Unfavourable (mild)
Cell Senescence	Radiation induces senescence; TP53 Arg/Arg favors apoptosis instead of senescence	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Isotretinoin and sun:** while taking isotretinoin, broad spectrum protector FPS 50 every day on the face, neck and hands, renewed every 2 h in the sun, and clothes or hat at long exposures. Genetic skin is protective, but isotretinoin thins the corneal layer and facilitates scalding. *Suggestion (not applied).*
2. **Skin examination:** with low genetic risk of melanoma, an annual skin examination with dermoscopy arrives. The monthly self-examination seeks signs that change (rule ABCDE). *Suggestion (not applied).*
3. **Medical radiation:** prefer ultrasound or magnetic resonance imaging when giving the same answer, ask if a TAC changes the decision before doing it, and save the record of the TAC already made. With ATM D1853N in homozygosity, this weighs a little more. *Suggestion (not applied).*
4. **Radon at home:** radon is the second cause of lung cancer. A test of 3 months on the lowest floor inhabited costs little; above 300 Bq/m³, ventilating or sealing the pavement. *Suggestion (not applied).*
5. **Thyroid:** in dental X-rays, accept the thyroid protection collar. Potassium iodide tablets are only taken as an indication of authorities in a nuclear emergency. *Suggestion (not applied).*
6. **Rapamycin and skin:** mTOR inhibitors are associated with less squamous cell carcinomas in transplant recipients; it is not a reason to change anything, but it is a point in favor. *Suggestion (not applied).*
7. **If one day you need radiotherapy:** tell the radiotherapist that has ATM D1853N in homozygosity and high loci load for toxicity, for planning to take this into account (the evidence is still weak). *Suggestion (not applied).*

Not recommended: solaria (cancerigen of group 1, without safe dose); TAC of whole body tracking without indication; exchange the sunscreen with "photoprotective" oral supplements.

10. Limitations

- The chip reads only the known variants of MC1R and some repair genes. High-risk rare mutations (CDKN2A, CDK4, BAP1, POT1, ATM causative) are not evaluated here.
- Radiosensitivity has few replicated genetic studies; there are no polygenic scores validated for it, and the loci load of radiotherapy toxicity joins organ studies and different outcomes.
- The two basal cell carcinoma scores disagree (low and typical percentiles); the most recent one uses MTAG and combines related traits.
- The risk of skin cancer depends mainly on accumulated sun exposure, scalding in childhood and number of signs; thyroid exposure in childhood.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location are read from the Imputed VCF. The published effect allele is compared with those alleles and, if necessary, converted to the complementary strand (only in non-palindromic SNPs). In palindromic SNPs, strand + (GWAS Catalog standard and the harmonized files of PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog: PGS002155 (Privé 2022), PGS001257 (Tanigawa 2022), PGS000079 and PGS000087 (Graff 2021), PGS000409 (Fritsche 2020), PGS000119 (Fritsche 2019), PGS003416 and PGS005209. Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** GWAS Catalog, ClinVar, dbSNP/Ensembl and reference literature: Raimondi 2008 (MC1R and melanoma), Law 2015 (GWAS of melanoma), Andreassen 2016 and Kerns 2020 (genetics of radiotherapy toxicity), Takahashi 2010 (FOXE1 and NKX2-1 in radiation-induced thyroid cancer) and the label of isotretinoin. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
MC1R (skin response to sun)	MC1R	rs1805007	C/C	chip	T	0	strand +	without the risk allele
MC1R (skin response to sun)	MC1R	rs1805008	C/C	chip	T	0	strand +	without the risk allele
MC1R (skin response to sun)	MC1R	rs1805009	G/G	chip	C	0	strand + (palindromic; assumed +)	without the risk allele
MC1R (skin response to sun)	MC1R	rs1805005	G/G	chip	T	0	strand +	without the risk allele
MC1R (skin response to sun)	MC1R	rs11547464	G/G	chip	A	0	strand +	without the risk allele
MC1R (skin response to sun)	MC1R	rs1805006	C/C	chip	A	0	strand +	without the risk allele
MC1R (skin response to sun)	MC1R	rs1110400	T/T	chip	C	0	strand +	without the risk allele
Pigmentation and sensitivity to sun	IRF4	rs12203592	C/C	chip	T	0	strand +	without the risk allele
Pigmentation and sensitivity to sun	SLC45A2	rs16891982	C/G	chip	C	1	strand + (palindromic; assumed +)	1 copy of the protective allele
Pigmentation and sensitivity to sun	TYR	rs1126809	G/G	imputation DR2=1.00	A	0	strand +	without the risk allele
Pigmentation and sensitivity to sun	TYR	rs1042602	C/A	chip	A	1	strand +	1 copy of risk allele
Pigmentation and sensitivity to sun	HERC2/OCA2	rs12913832	A/G	chip	G	1	strand +	1 copy of risk allele
Pigmentation and sensitivity to sun	ASIP	rs4911414	G/G	chip	T	0	strand +	without the risk allele
DNA repair and ionizing radiation	ATM	rs1801516	A/A	chip	A	2	strand +	2 risk allele copies
DNA repair and ionizing radiation	XRCC1	rs25487	C/C	chip	A	0	complement (strand)	without the risk allele
DNA repair and ionizing radiation	XRCC1	rs1799782	G/G	chip	T	0	complement (strand)	without the risk allele
DNA repair and ionizing radiation	XRCC3	rs861539	G/G	chip	T	0	complement (strand)	without the risk allele
DNA repair and ionizing radiation	ERCC2 (XPD)	rs13181	T/T	chip	C	0	complement (strand)	without the risk allele
DNA repair and ionizing radiation	TP53	rs1042522	C/C	imputation DR2=0.97	G	0	strand + (palindromic; assumed +)	0 copy of the effect allele
DNA repair and ionizing radiation	TGFB1	rs1800469	A/G	chip	T	1	complement (strand)	1 copy of risk allele
Thyroid and radiation	FOXE1	rs965513	A/G	chip	A	1	strand +	1 copy of risk allele
Thyroid and radiation	NKX2-1/MBIP	rs944289	C/T	chip	T	1	strand +	1 copy of risk allele